CLASSIFICATION / ODD PAIR

Odd Word, Odd Pair, Odd Letter-Cluster and Odd Number

OAVS TGT | Reasoning (Verbal) | Topic 7 of the syllabus list

Syllabus line (Reasoning, 30 Marks): “The Verbal Reasoning topics like Series, Ranks, Direction, Arrangement, Coding, Decoding, Analogy and **Classification/Odd Pair**, Syllogism and Statement Conclusion.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). All 11 classification questions found in these papers are solved here. English 2019 had none.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	Counts, patterns, priority, index of all 11 PYQs
2	Foundation	What ‘odd’ means, the 4-step method, property checklist
3	Odd Word	Category and meaning (2 PYQs)
4	Odd Word-Pair	Place–thing, place–worker, study–subject, synonyms (4 PYQs)
5	Letter and Alphanumeric Clusters	Odd clusters by shift pattern (4 PYQs)
6	Jumbled Words	Unscramble then classify (1 PYQ)
7	Odd Number (expected)	Primes, squares, cubes, multiples, digit rules
8	Shortcuts, Traps and Scope	Speed methods; what belongs to Series or Analogy
9	Practice Set	22 original questions with answer key
10	Quick Revision	One-page cheat sheet

SCOPE: Questions that ask ‘which letter-cluster / number does NOT belong to the series’ (Maths 2024 Q80, Q99; Science 2024 Q73; SST 2023 Q11) are series questions and were solved in the Series PDF. ‘Same pattern’ triads and ‘A is related to B’ questions are in the Analogy PDF.

CHAPTER 1

PYQ Analysis

Classification: about 2 questions per paper

Paper	Q. numbers	Count	What was asked
English 2019	none	0	—
English 2022	38, 51, 58, 59	4	Odd word (2); jumbled words; alphanumeric clusters
Mathematics 2024	72, 85	2	Odd pair of letters; odd three-letter cluster
Science 2024	72, 87, 89	3	Odd word-pair (2); odd letter-cluster
Social Studies 2023	5, 29	2	Odd word-pair (study : subject; synonyms)
Total		**11**	**Found in four of the five papers**

Pattern	PYQs	Priority
Odd word-pair (find the pair whose relationship differs)	4	★★★ Must
Odd letter-cluster by shift pattern (including pairs and letter + number)	4	★★★ Must
Odd word (category or meaning)	2	★★ High
Jumbled words, unscramble and classify	1	★★ High
Expected, not yet asked: odd number (primes, squares, cubes, multiples)	0	★ Expected

PYQ INSIGHT: **Four of 11 are word pairs** (Science 2024 Q72, Q89; SST 2023 Q5, Q29). In every one the odd pair has a **different relationship** from the other three: place–worker vs place–thing; study–subject; synonym vs not synonym. Name each pair's relationship and the odd one shows up.

PYQ INSIGHT: **Letter-cluster odd-ones all use shifts**: the three 'normal' clusters follow one pattern of letter-to-letter steps and the odd one breaks it by 1 place (Maths 2024 Q72: +5 instead of +6; Q85: +9, −6 instead of +10, −5; Science 2024 Q87: +2, +2 instead of +2, +3).

Index of the 11 PYQs

Paper / Q	Options (short)	Odd one	See
Eng 2022 Q38	Gleeful, Jovial, Grinning, Cheerless	Cheerless	3.1
Eng 2022 Q58	DuckDuckGo, Yandex, Pinterest, Google	Pinterest	3.2
Science 2024 Q72	Theatre–Actors, School–Teachers, Garage–Cars, Laboratory–Scientists	Garage–Cars	4.1
Science 2024 Q89	Garage–Tools, Library–Books, Kitchen–Utensils, Classroom–Guns	Classroom–Guns	4.1
SST 2023 Q5	Palaeography : Writings; Seismology : Earthquakes; Ornithology : Birds; Nidology : Moon	Nidology : Moon	4.2
SST 2023 Q29	Hazardous : Perilous; Tedious : Panicked; Sparkling : Radiant; Indolent : Slothful	Tedious : Panicked	4.3
Maths 2024 Q72	IE-HK, RN-QT, LH-KM, OK-NQ	LH-KM	5.1
Maths 2024 Q85	EOJ, KUP, PYS, HRM	PYS	5.2
Science 2024 Q87	XZB, SUX, BDG, GIL	XZB	5.3
Eng 2022 Q59	12L, 18R, 7G, 16Q	16Q	5.4
Eng 2022 Q51	LECIBYC, ECOROTS, REELANOPA, SAPESGERNE	SAPESGERNE	6.1

CHAPTER 2

Foundation

What ‘odd’ means and how to find it

2.1 The question types

Type	Typical wording	You find
Odd word	Four words; three are alike in some manner and one is different	The word that does not share the common property
Odd word-pair	Select the word-pair in which the two words share a different relationship from the rest	The pair with a different relationship
Odd letter-cluster / pair	Three of the four letter-clusters are alike in a certain way; which does not belong?	The cluster that breaks the shift pattern
Odd alphanumeric	Four alphanumeric-clusters; select the different one	The cluster that breaks the letter–number rule
Jumbled words	Four letter-clusters; three are alike, one different	Unscramble each, then classify

2.2 Four-step method

Step	Do this
1	Read all four. Do **not** assume the first three are the group.
2	Find a property shared by **any three** (category, meaning, relationship, shift pattern, spelling) and test it on the fourth.
3	If two options both look odd, find a **second property** and pick the one the property singles out clearly.
4	Re-check by naming why the other three belong together in one sentence. If you cannot, the property is wrong.

2.3 Property checklist

Items are…	Check for	Example
Words	Category (animals, vehicles, search engines), meaning (synonym / antonym), function, material	Gleeful, Jovial, Grinning | Cheerless
Word pairs	Relation: place–worker, place–thing, study–subject, tool–user, synonym, antonym, part–whole	Garage–Cars vs place–worker
Letter clusters	Shift from letter to letter; positions; vowels (only if the question allows)	XZB: +2, +2 vs the rest +2, +3
Alphanumeric	Number = letter position, or letter and number relationship	12L: 12 = L’s position
Numbers	Prime, square, cube, multiple, odd / even, digit sum	9, 16, 25, 36, 48 → 48

SHORTCUT: The question stem sometimes **rules out** a property: Maths 2024 Q72 says ‘the odd one is not based on the number of consonants / vowels or their position’. Do not waste time on vowel-counting there; use the shifts.

CHAPTER 3

Odd Word

Category and meaning: 2 PYQs

3.1 Odd by meaning

PYQ Eng 2022 Q38 Opposite among synonyms

Q. Four words have been given, out of which three are alike in some manner and one is different. A. Gleeful B. Jovial C. Grinning D. Cheerless

Step 1. Gleeful, jovial and grinning all describe being happy or cheerful.
Step 2. Cheerless means gloomy or sad: the opposite of the other three.
Answer: Cheerless (Option D)

3.2 Odd by category

PYQ Eng 2022 Q58 Search engines

Q. Select the word that is different. A. DuckDuckGo B. Yandex C. Pinterest D. Google

Step 1. DuckDuckGo, Yandex and Google are web search engines.
Step 2. Pinterest is an image-sharing and discovery (social) platform, not primarily a search engine.
Answer: Pinterest (Option C)

SHORTCUT: **Category names to know** (they are the usual hidden property): search engines (Google, Bing, Yandex, DuckDuckGo), social media (Facebook, Instagram, Pinterest), browsers (Chrome, Firefox), mammals vs birds vs reptiles, vegetables vs fruits, rivers vs mountains, professions vs places.

3.3 Category tests (expected)

Group	Test	Example odd one
Animals	Mammal, bird, reptile, fish, insect	Bat among birds
Food	Fruit vs vegetable vs grain	Potato among fruits
Geography	River, mountain, state, capital	Himalaya among rivers
People and places	Profession vs place vs object	Hospital among professions
Technology	Search engine, social media, browser, language	Chrome among social media

CHAPTER 4

Odd Word-Pair

The pair whose relationship differs: 4 PYQs

4.1 Place and what is there (Science 2024 Q72, Q89)

PYQ Science 2024 Q72 Place–worker vs place–thing

Q. Three of the following word-pairs are alike in some manner. Which pair does not belong? 1. Theatre–Actors 2. School–Teachers 3. Garage–Cars 4. Laboratory–Scientists

Step 1. Theatre–Actors, School–Teachers, Laboratory–Scientists: a **place** and the **people who work there**.
Step 2. Garage–Cars: a place and the **things kept** there, not the workers.
Answer: Garage–Cars (Option 3)

PYQ Science 2024 Q89 Place–things usually kept vs not

Q. Three of the following word-pairs are alike. Which does not belong? 1. Garage–Tools 2. Library–Books 3. Kitchen–Utensils 4. Classroom–Guns

Step 1. Garage–Tools, Library–Books, Kitchen–Utensils: a place and the objects normally **kept or used** there.
Step 2. Guns are not part of a classroom: the relation fails.
Answer: Classroom–Guns (Option 4)

TRAP: Q72 and Q89 both start with ‘Garage’ but have **different** odd answers: in Q72 the odd one is the pair that is **not** worker-based; in Q89 it is the pair that is not **normally kept there**. Decide the relation from the other three, not from the first word.

4.2 Study and subject (SST 2023 Q5)

PYQ SST 2023 Q5 Science : what it studies

Q. Select the word-pair that shares a different relationship. 1. Palaeography : Writings 2. Seismology : Earthquakes 3. Ornithology : Birds 4. Nidology : Moon

Step 1. The pattern is **study of** and the thing it studies: palaeography studies ancient writings, seismology earthquakes, ornithology birds.
Step 2. Nidology is the study of **nests**, not the Moon.
Answer: Nidology : Moon (Option 4)

Study	Subject	Study	Subject
Palaeography	ancient writings	Seismology	earthquakes
Ornithology	birds	Nidology	nests
Cardiology	heart	Entomology	insects
Horology	time / clocks	Ichthyology	fish

4.3 Synonym vs not (SST 2023 Q29)

PYQ SST 2023 Q29 Synonym pairs

Q. Select the word-pair that shares a different relationship. 1. Hazardous : Perilous 2. Tedious : Panicked 3. Sparkling : Radiant 4. Indolent : Slothful

Step 1. Hazardous–perilous, sparkling–radiant, indolent–slothful are **synonyms**.
Step 2. Tedious (boring) and panicked (frightened) have no shared meaning.
Answer: Tedious : Panicked (Option 2)

SHORTCUT: Pair-relation list to test: synonym, antonym, place–worker, place–thing, study–subject, tool–user, part–whole, young–adult, cause–effect. Name each pair’s relation; the one with a different name is odd.

CHAPTER 5

Letter and Alphanumeric Clusters

Find the one that breaks the pattern: 4 PYQs

Method

Step	Do this
1	Convert every cluster to alphabet positions (A = 1 … Z = 26).
2	Write the **step from letter to letter** inside each cluster (+k or −k, with wrap-around).
3	Three clusters share the same steps; the odd one differs. For pairs (such as IE-HK), compare each letter of the first with the same position of the second.

5.1 Pairs of letters (Maths 2024 Q72)

PYQ Maths 2024 Q72 Pairs, shift per letter

Q. Three of the letter-cluster pairs are alike. Which does not belong? (The odd one is not based on the number of consonants / vowels or their position.) A. IE-HK B. RN-QT C. LH-KM D. OK-NQ

Step 1. Compare the first letters: I → H, R → Q, L → K, O → N: −1 in every pair.
Step 2. Second letters: E → K (+6), N → T (+6), H → M (+5), K → Q (+6).
Step 3. LH-KM has +5 where the others have +6.
Answer: LH-KM (Option C)

Pair	Position change: first letter	second letter
IE–HK	I → H: -1	E → K: +6
RN–QT	R → Q: -1	N → T: +6
LH–KM	L → K: -1	H → M: +5
OK–NQ	O → N: -1	K → Q: +6

5.2 Three-letter clusters (Maths 2024 Q85)

PYQ Maths 2024 Q85 Steps inside the cluster

Q. Three of the four are alike. Which does not belong? A. EOJ B. KUP C. PYS D. HRM

Step 1. EOJ: E (5) → O (15) is +10; O → J (10) is −5. KUP: K (11) → U (21) is +10; U → P (16) is −5. HRM: H (8) → R (18) is +10; R → M (13) is −5.
Step 2. PYS: P (16) → Y (25) is +9; Y → S (19) is −6. This breaks the pattern.
Answer: PYS (Option C)

Cluster	Positions	Shifts (each letter to the next)
EOJ	5 15 10	+10, -5
KUP	11 21 16	+10, -5
PYS	16 25 19	+9, -6
HRM	8 18 13	+10, -5

5.3 Steps with a wrap-around (Science 2024 Q87)

PYQ Science 2024 Q87 +2, +3 pattern

Q. Three of the letter-clusters are alike. Which does not belong? 1. XZB 2. SUX 3. BDG 4. GIL

Step 1. SUX: S → U +2, U → X +3. BDG: B → D +2, D → G +3. GIL: G → I +2, I → L +3.
Step 2. XZB: X (24) → Z (26) is +2, Z (26) → B (2) is +2 (wraps past Z). The second step is +2, not +3.
Answer: XZB (Option 1)

Cluster	Positions	Shifts (each letter to the next)
XZB	24 26 2	+2, +2
SUX	19 21 24	+2, +3
BDG	2 4 7	+2, +3
GIL	7 9 12	+2, +3

TRAP: XZB has the same first step (+2) as the others, so checking only the first step would not find it. Compute **both** steps, and remember Z → B is +2, not −24.

5.4 Letter with a number (Eng 2022 Q59)

PYQ Eng 2022 Q59 Number equals letter position

Q. Four alphanumeric-clusters; select the different one. A. 12L B. 18R C. 7G D. 16Q

Step 1. L is the 12th letter, R the 18th, G the 7th: the number equals the position of the letter.
Step 2. Q is the 17th letter, but the cluster shows 16.
Answer: 16Q (Option D)

SHORTCUT: For alphanumeric clusters try three tests in turn: number = position of the letter; number = position from the end (27 − number); number is a multiple or square related to the letter. The first one solved this PYQ.

CHAPTER 6

Jumbled Words

Unscramble, then classify: 1 PYQ

PYQ Eng 2022 Q51 Anagrams of transport

Q. Four letter-clusters; three are alike and one is different. A. LECIBYC B. ECOROTS C. REELANOPA D. SAPESGERNE

Step 1. Unscramble each: LECIBYC = BICYCLE; ECOROTS = SCOOTER; REELANOPA = AEROPLANE.
Step 2. SAPESGERNE unscrambles to PASSENGER (the printed cluster has an extra E, a slip in the source).
Step 3. Bicycle, scooter and aeroplane are vehicles; a passenger is a person.
Answer: SAPESGERNE (Option D)

SHORTCUT: **How to unscramble quickly:** start from the most distinctive letters (Y, C, B → BICYCLE; P, L, N → AEROPLANE), match length first, then letters. If the odd cluster is much longer or shorter, it is often the odd one (SAPESGERNE has 10 letters).

Likely alternative: same-category check

Scrambled	Word	Category
LEEPTHAN	ELEPHANT	animal
RIGET	TIGER	animal
HRSOE	HORSE	animal
BLEAT	TABLE	furniture (odd)

CHAPTER 7

Odd Number (expected)

Not yet asked in these papers; likely in other papers

Number group	Property	Odd one example
Primes	Only divisible by 1 and itself	11, 13, 17, **21**, 23 → 21 (3 × 7)
Perfect squares	n^2: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100	9, 16, 25, 36, **48** → 48
Perfect cubes	n^3: 1, 8, 27, 64, 125, 216	8, 27, 64, **100**, 125 → 100
Multiples	Multiples of the same number	12, 18, 24, **27**, 30 → 27 (not a multiple of 6)
Odd / even	Parity	2, 4, 6, **9**, 12 → 9
Digit sum / digits	Same digit sum or divisible by 9	18, 27, 36, **38** → 38

SHORTCUT: For numbers, check **prime, square, cube, multiple** in that order; usually one of them separates the odd number in a few seconds.

CHAPTER 8

Shortcuts, Traps and Scope

Speed methods and what is covered elsewhere

- **Name the relation or property** of each option in one word.
- **Pairs:** compare the first letters, then the second letters of the pairs.
- **Clusters:** compute **both** letter-to-letter steps; one step may match.
- **Wrap-around:** Z → B is +2.
- **Read the stem:** it may rule out a property (vowels, consonants, position).
- **Alphanumeric:** test 'number = letter position' first.
- **Jumbled words:** unscramble the most distinctive letters first.

Trap	How to avoid it
Assuming the first three form the group	Test properties on any three of the four.
Two plausible odd answers	Find a second property, or the stricter relationship (place–worker vs place–thing).
Checking one step of a cluster	XZB matched the first step of the others.
Same first word, different pattern	Garage–Cars (Q72) and Classroom–Guns (Q89) are odd for different reasons.
Treating a longer jumbled word as one of the group	Unscramble before classifying; the odd one is often a different category.

Scope check

PYQ	Where it belongs
Maths 2024 Q80, Q99; Science 2024 Q73; SST 2023 Q11	'Incorrect / does not belong to the series': solved in the Series PDF
Maths 2024 Q90, Q100; Science 2024 Q99; SST 2023 Q2, Q10	'Same pattern' triads and relationship questions: Analogy PDF

CHAPTER 9

Practice Set

22 original questions; answers on the next page

Q	Question
1	Mango, Banana, Potato, Apple. Select the odd one.
2	Eagle, Sparrow, Bat, Parrot. Select the odd one.
3	Ganga, Yamuna, Godavari, Himalaya. Select the odd one.
4	Doctor, Engineer, Hospital, Teacher. Select the odd one.
5	Happy, Joyful, Cheerful, Sad. Select the odd one.
6	Cobra, Viper, Python, Crocodile. Select the odd one.
7	Facebook, Instagram, Twitter, Chrome. Select the odd one.
8	Hindi, Odia, Bengali, Rupee. Select the odd one.
9	Which pair is different? 1. Pilot–Aeroplane 2. Driver–Bus 3. Sailor–Ship 4. Doctor–Stethoscope
10	Which pair is different? 1. Cow–Calf 2. Dog–Puppy 3. Cat–Kitten 4. Hen–Egg
11	Which pair is different? 1. Hot–Cold 2. Big–Small 3. Rich–Poor 4. Quick–Fast
12	Which pair is different? 1. Potter–Clay 2. Carpenter–Wood 3. Sculptor–Stone 4. Farmer–Tractor
13	Which letter-cluster does not belong? 1. ACE 2. FHJ 3. KMO 4. PRU
14	Which letter-cluster pair does not belong? 1. DA-CG 2. JF-IL 3. NK-MP 4. PM-OS
15	Which alphanumeric cluster does not belong? 1. 4D 2. 9I 3. 14N 4. 20U
16	Which letter-cluster does not belong? 1. BEH 2. KNQ 3. PSV 4. TWY
17	Which jumbled word is different? 1. LEEPTHAN 2. RIGET 3. HRSOE 4. BLEAT
18	Which number does not belong? 9, 16, 25, 36, 48
19	Which number does not belong? 11, 13, 17, 21, 23
20	Which number does not belong? 8, 27, 64, 100, 125
21	Which pair is different? 1. Seismology–Earthquakes 2. Cardiology–Heart 3. Ornithology–Birds 4. Horology–Rainfall
22	Which pair is different? 1. Hazardous–Perilous 2. Brave–Fearless 3. Calm–Agitated 4. Large–Huge

Answer key with logic

Q	Answer	Logic
1	Potato	vegetable; the rest are fruits
2	Bat	mammal; the rest are birds
3	Himalaya	mountain; the rest are rivers
4	Hospital	place; the rest are professions
5	Sad	opposite of the other three
6	Crocodile	not a snake
7	Chrome	browser; the rest are social media
8	Rupee	currency; the rest are languages
9	Doctor–Stethoscope	the rest are person and vehicle operated; this is person and tool
10	Hen–Egg	the rest are animal and young one
11	Quick–Fast	synonyms; the rest are antonyms
12	Farmer–Tractor	worker and tool; the rest are worker and raw material
13	PRU	ACE, FHJ, KMO are +2, +2; PRU is +2, +3
14	NK-MP	first −1, second +6 in the rest; NK-MP has K → P +5
15	20U	number = letter position (D 4, I 9, N 14); U is the 21st, not 20
16	TWY	BEH, KNQ, PSV are +3, +3; TWY is +3, +2
17	BLEAT	TABLE (furniture); the rest are animals (ELEPHANT, TIGER, HORSE)
18	48	the rest are perfect squares
19	21	the rest are primes (21 = 3 × 7)
20	100	the rest are perfect cubes
21	Horology–Rainfall	study and subject; horology studies time, not rainfall
22	Calm–Agitated	antonyms; the rest are synonyms

CHAPTER 10

Quick Revision

One page for the last day

Type	Method	PYQ check
Odd word	Find the shared category or meaning of three	Cheerless (opposite); Pinterest (not a search engine)
Odd word-pair	Name each pair's relation	Garage–Cars (thing); Classroom–Guns; Nidology : Moon; Tedious : Panicked
Odd letter pair	Compare positions in each pair	LH-KM: +5 instead of +6
Odd letter cluster	Both letter-to-letter steps	PYS (+9, −6); XZB (+2, +2)
Odd alphanumeric	Number = letter position?	16Q (Q = 17)
Jumbled words	Unscramble, then classify	BICYCLE, SCOOTER, AEROPLANE | PASSENGER
Odd number	Prime, square, cube, multiple	48 among squares; 21 among primes

ROUTINE: **Exam-day routine:** name the common property of three, check the fourth; for clusters, write both steps; use the stem's hints. 25 seconds per question.